Experiment - 1

AIM :- To design an ER diagram for an Indian Ecommerce platform containing entities : Customer, Product Order, Order Item, Seller, Category, Payment delivery and address.

THEORY :- An Entity Relationship model is a conceptual representation of data used in a database design. It describes entities, their attributes and relationship among them.

KEY COMPONENTS :-

i) Entity - Real World Objects.

ii) Primary Key - Unique identities of an identity, cannot be null or empty or have duplicate entities.

iii) Relationship - describes how two or more entities are connected.

iv) Cardinality - how many instances of an entity can be associated with another. Types include such as one to one, one to many, many to many and even many to one.

Ex :- Customer 1 - N order (one to many)

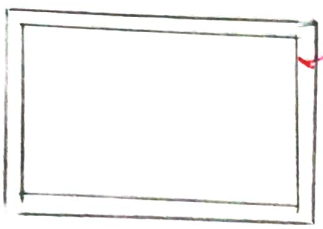
v) Weak Entity - Cannot be uniquely identified using its own attributes and depends on another entity.

eg :- order depends on order Item.

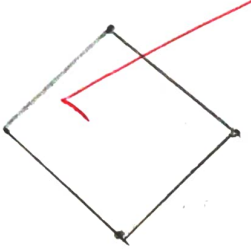
Notations Used :-



→ Strong Entity



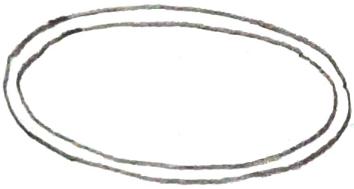
→ Weak Entity



→ Relationship



→ Attribute



→ Multi valued Attribute

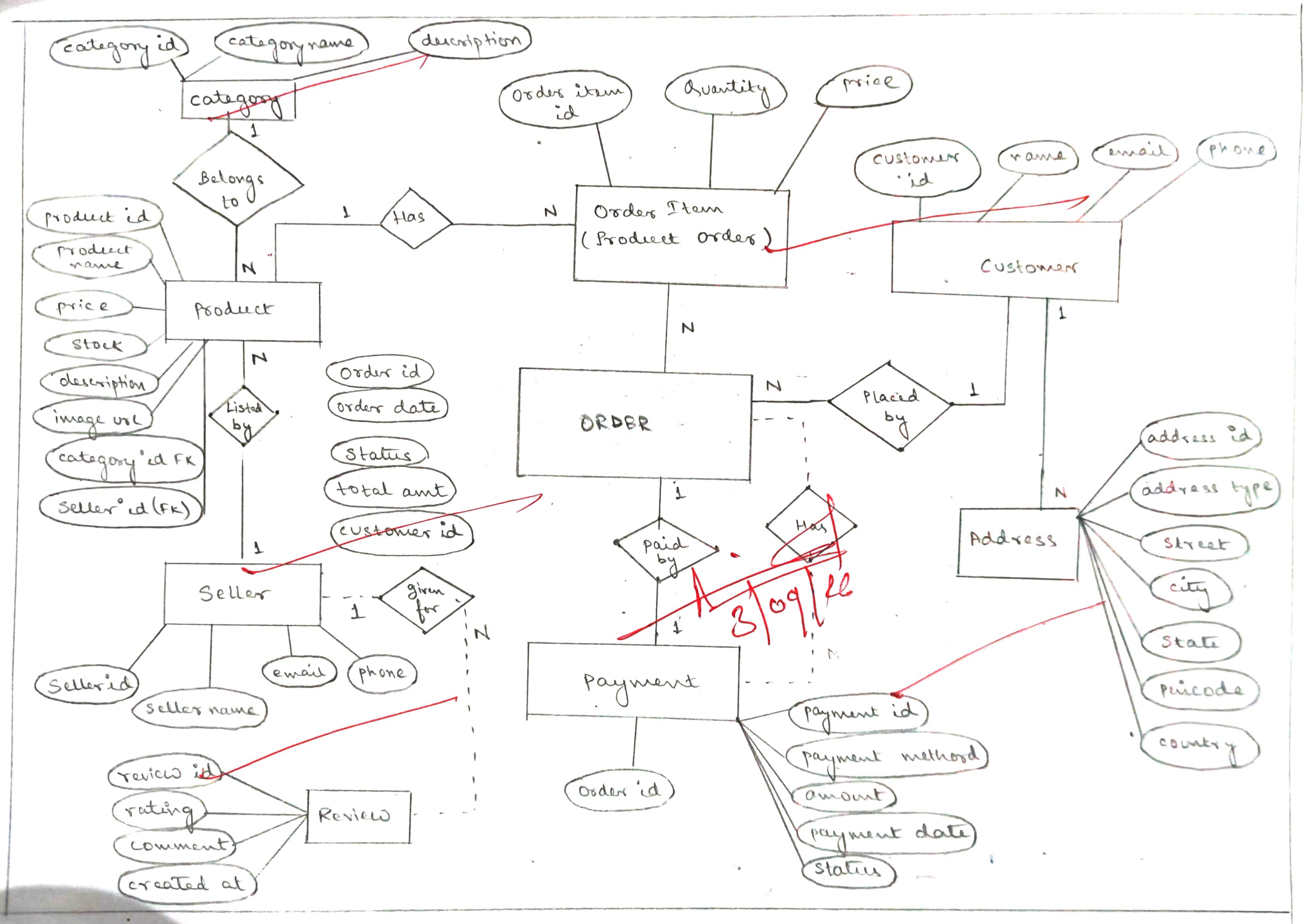


→ Composite Attribute



→ Key Attribute

- - - - - - → Partial Key


category id
category name
description
category
1
Belongs to
N
product id
product name
price
stock
description
image url
category id FK
Seller id (FK)
Product
N
Listed by
1
Seller
Seller id
Seller name
email
phone
1
given for
N
review id
rating
comment
created at
Review
1
Has
N
Order item id
Quantity
price
Order Item (Product order)
N
ORDER
Order id
order date
Status
total amt
customer id
1
paid by
1
Payment
Order id
payment id
payment method
amount
payment date
status
N
Placed by
1
customer id
name
email
phone
Customer
1
N
Address
address id
address type
street
city
state
pincode
country
Has
3/09/22

OBSERVATIONS :- The ER diagram successfully models the major entities and relationship of an e-commerce platform. Primary and Foreign Keys establish entity connections, while cardinalities defines how entities participate in relationship. The order Item entity resolves the many to many relationship between orders and products. The diagram provides a structure used formation for converting the conceptual model into normalized relational database tables.

RESULT :- The ER diagram for the e-commerce platform was successfully designed, identifying the major entities, their attributes, primary keys and relationship and even cardinalities. The diagram provides a clear conceptual model that can be converted into relationship related database tables and implemented using SQL.